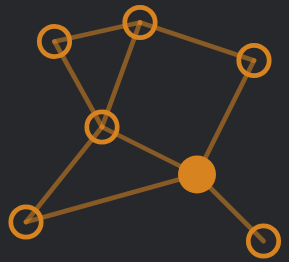
MidChain DeFi Conference

Metric-Based Voting

Designing Rules for Ecosystem Resource Allocation

…and first: food security on a blockchain



Nimrod Talmon

Ben-Gurion University of the Negev · October 26, 2026

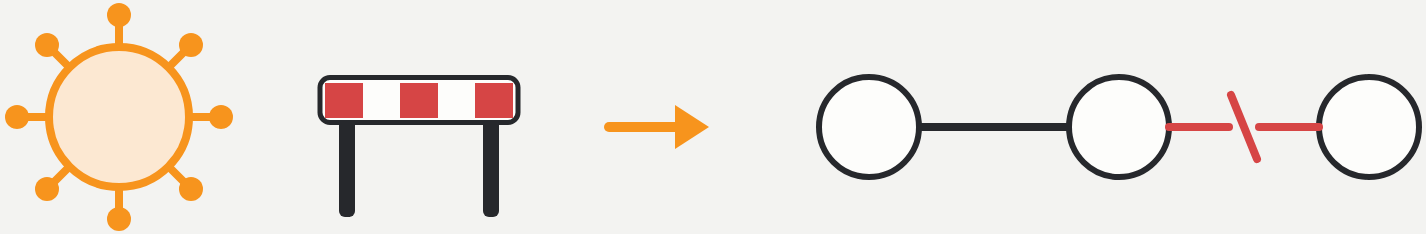
Part 1

Food security on a blockchain

with Bar Hoter, Moran Koren, Dorit Nitzan, Stav Shapira (+ Adir Even and Max Gluhov)

Supported by the Israeli Ministry of Science and Technology

Pandemics and wars break food supply chains





Also today: 1 in 4 households in Israel is food insecure

National Insurance Institute of Israel, Food Security Report 2024 (published 2025)

Goal: see the whole food system, live

1 · Record

every hand-off,
on-chain

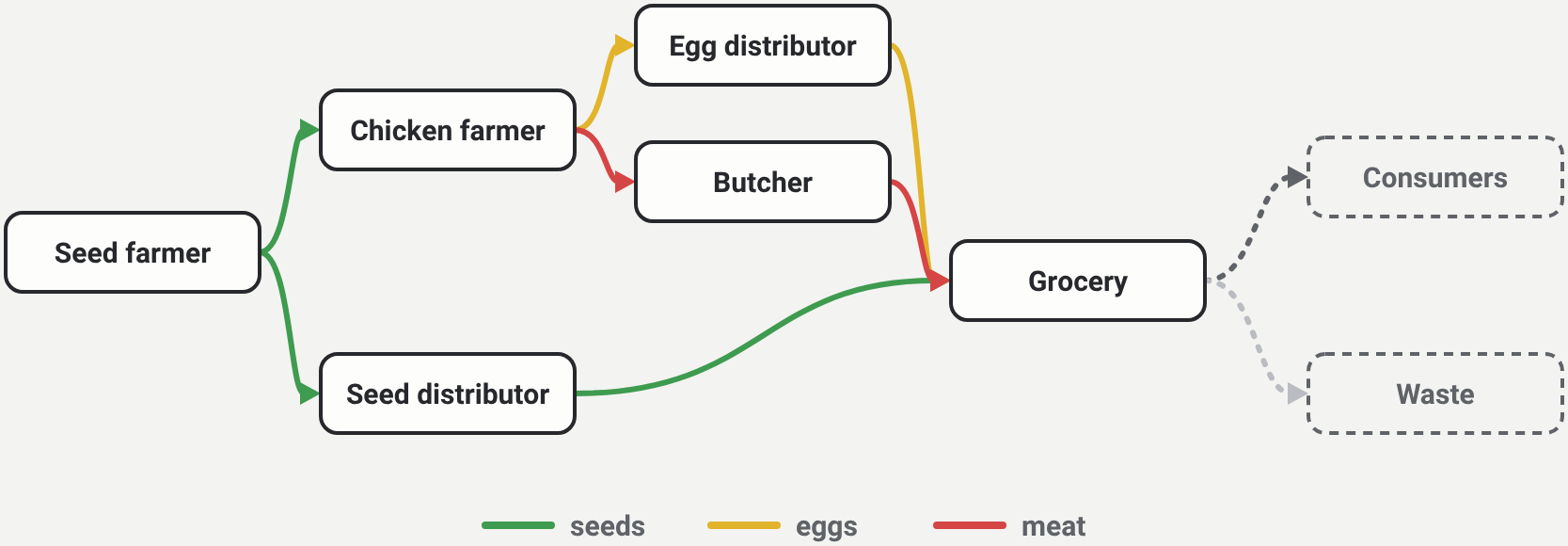
→

2 · Ask

How long will it last?
What can we still make?
What if a source is cut?

1 · Record

Food moves along a network of hand-offs

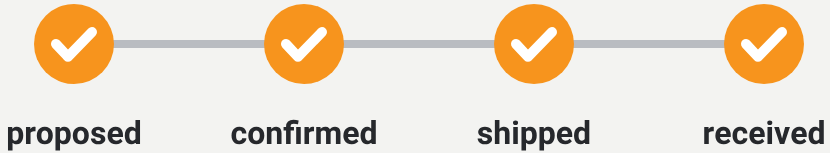


1 · Record

Every hand-off is a record

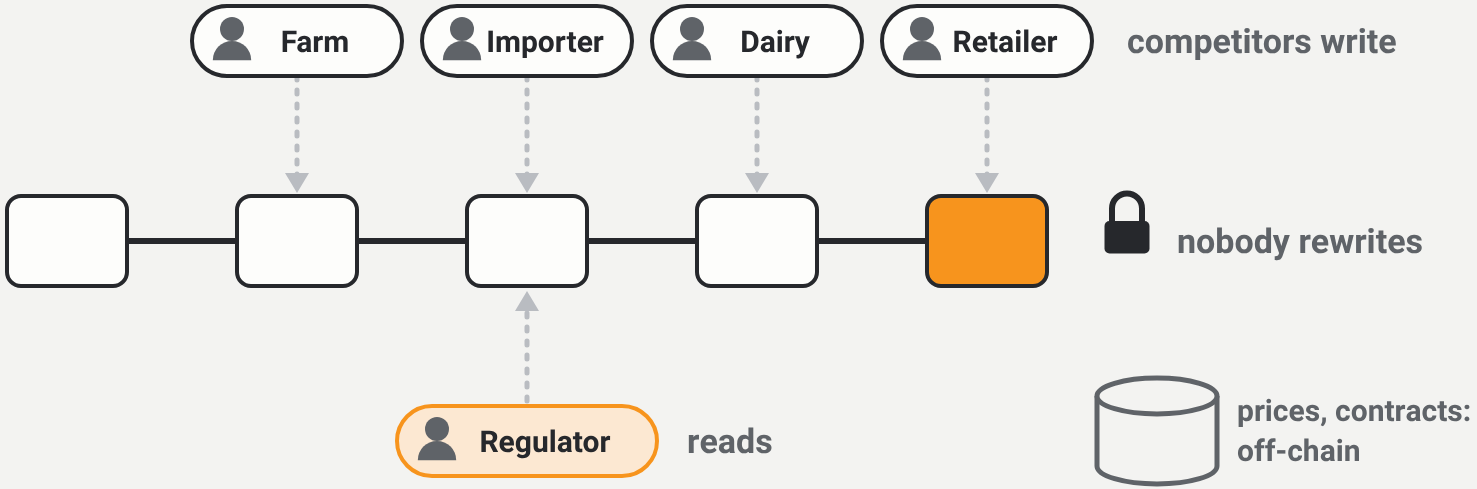
seller	**Dairy farm**
buyer	**Grocery**
batch	**#4417**
date	**March 3**
amount	**200 liters**

signed by both sides, on-chain



1 · Record

Why on-chain, for food?



And the Ministry of Science asked for it

2 · Ask

How long will the milk last?

in stock

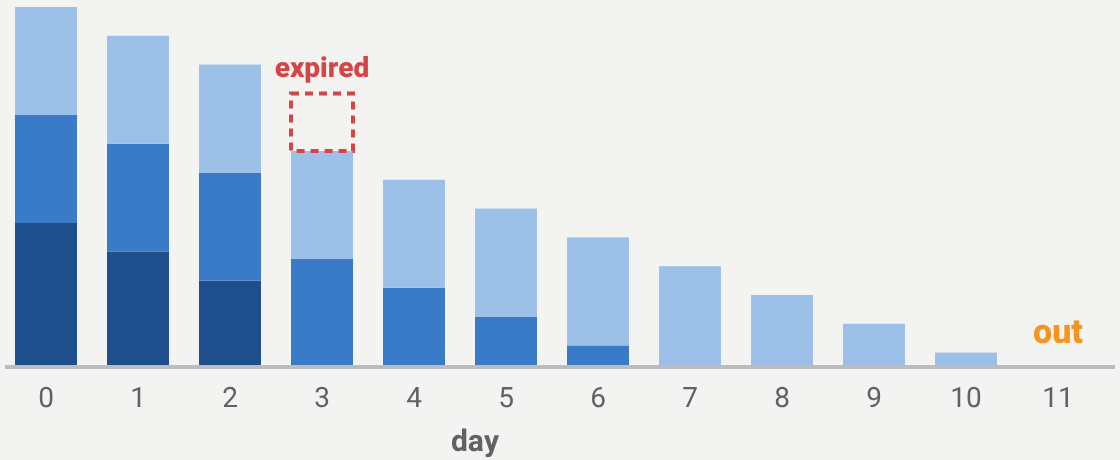
A — **40 liters**
good until day 3

B — **30 liters**
good until day 7

C — **30 liters**
good until day 13

we drink 8 a day

liters left, day by day



Expiry counts: out on day 11, not day 13

2 · Ask

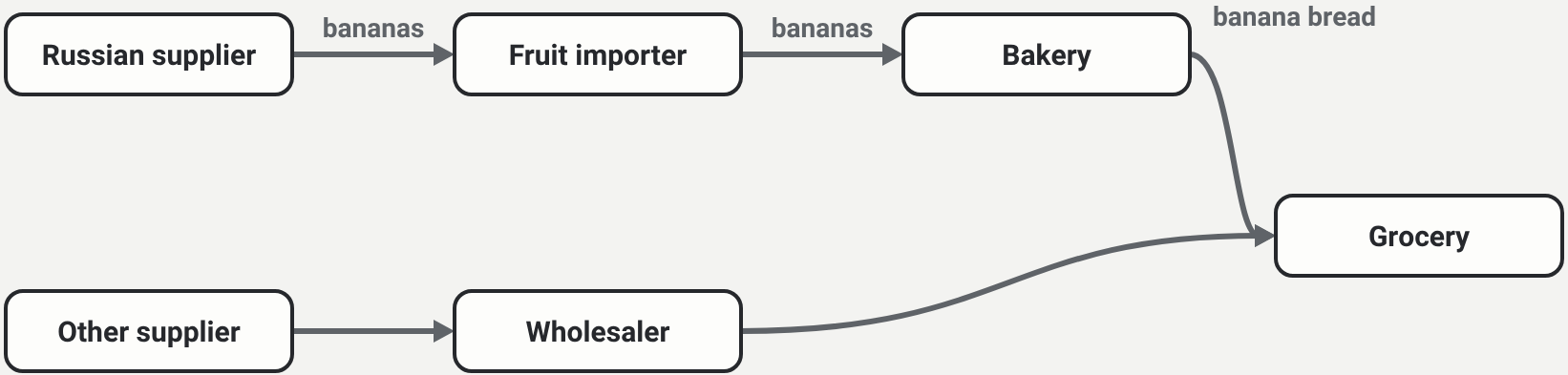
What can each producer still make?



The scarcest input decides

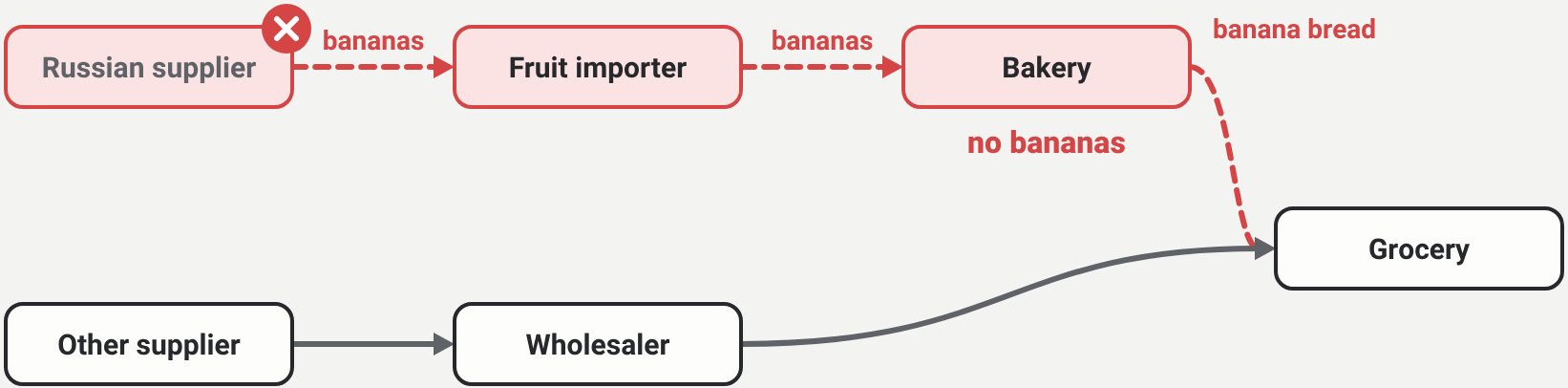
2 · Ask

What if we lose Russia?



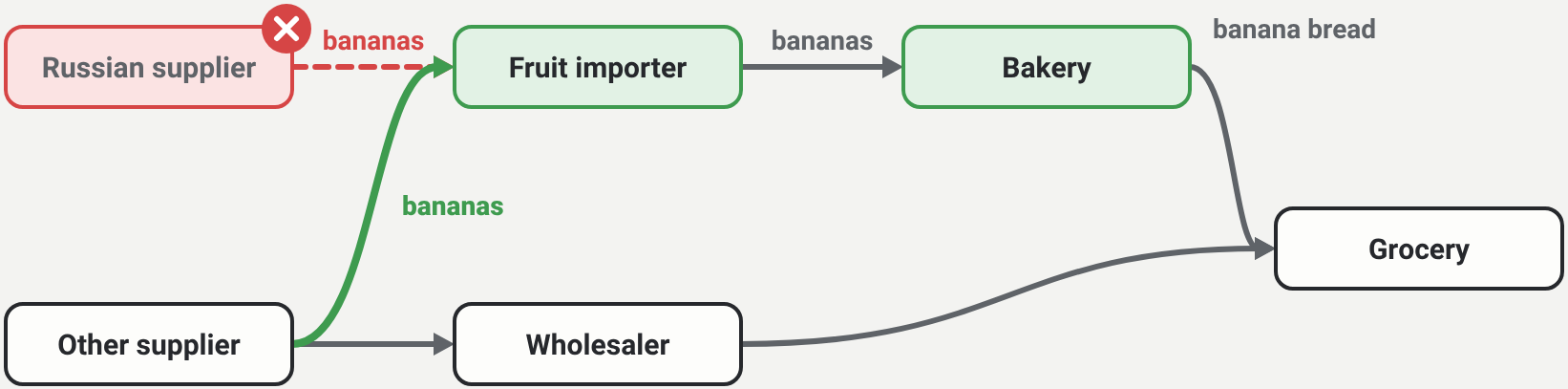
2 · Ask

What if we lose Russia?



2 · Ask

What if we lose Russia?



Bananas from elsewhere: the bakery is covered

Where it stands

Prototype

Ethereum testnet

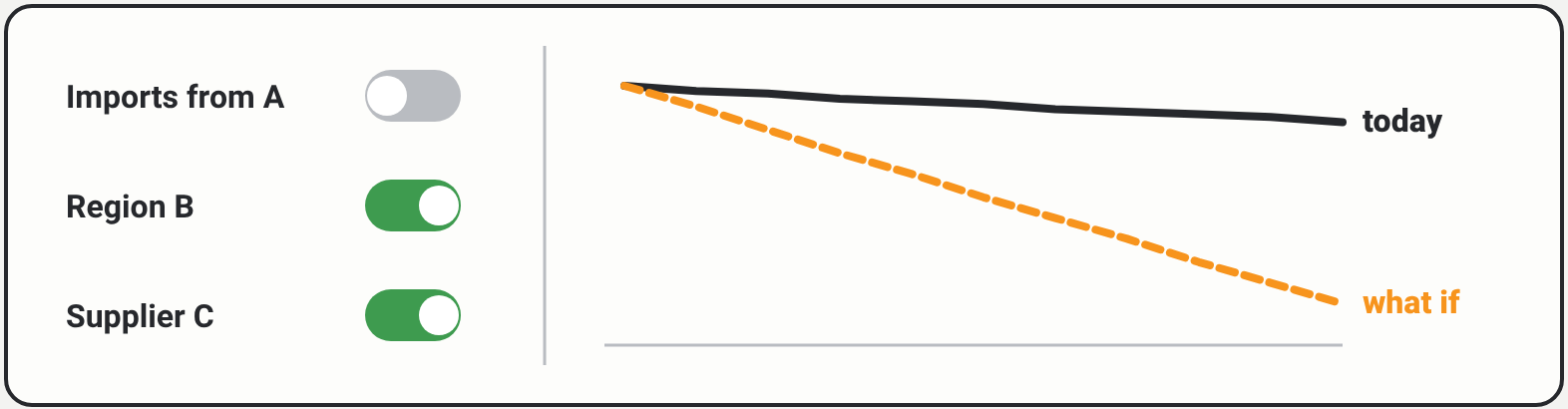
Data

Synthetic, for now

Dashboards

Nutrition and flows

Now: a real BI with a what-if simulator



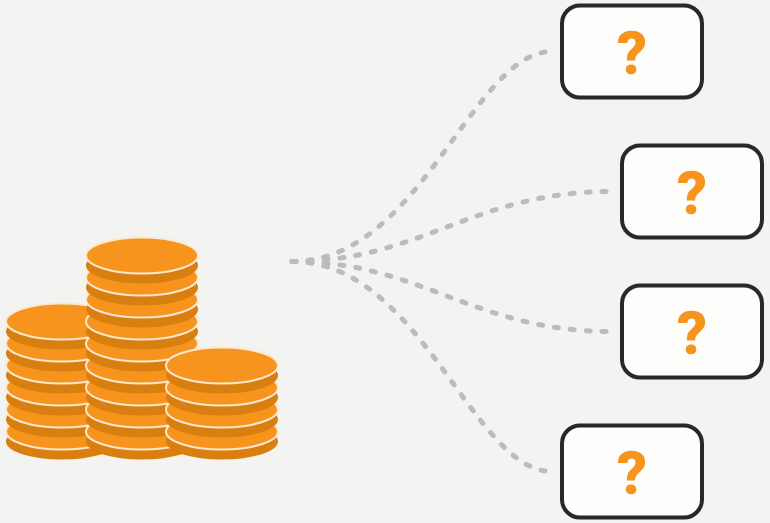
Interested? Let's talk, and stay in touch for updates

Part 2

Metric-based voting

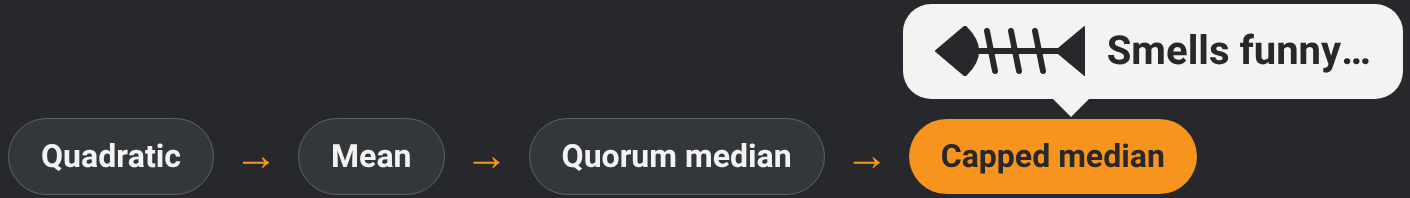
with Eyal Briman

A pot of money, many projects. Who gets what?



Optimism Retro Funding

Over $100M in four rounds, and a new voting rule every round



Briman, Talmon, Kreitenweis, Idrees 2025

Which rule is the right one?

This is what social choice studies



We looked at Optimism's rules

Found	The median rules invite strategic voting
Suggested	Moving phantoms: artificial voters so that lying never pays



Freeman, Pennock, Peters, Wortman Vaughan 2019 · Briman, Talmon, Kreitenweis, Idrees 2025

Round 4: vote on metrics, not on projects

108	16	230	10M
voters	metrics	projects	OP

Voters weighted metrics; onchain data scored the projects

Not only crypto

OECD Better Life Index

Rank countries: weight 11 topics

ERC

Rank grants: ambition, feasibility, PI



The model

Metric-based voting

How it works

Metrics

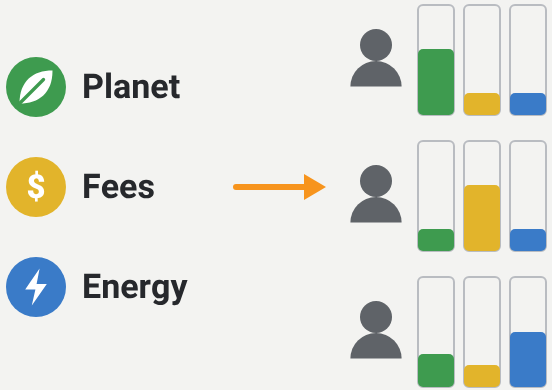
what counts

- Planet
- Fees
- Energy

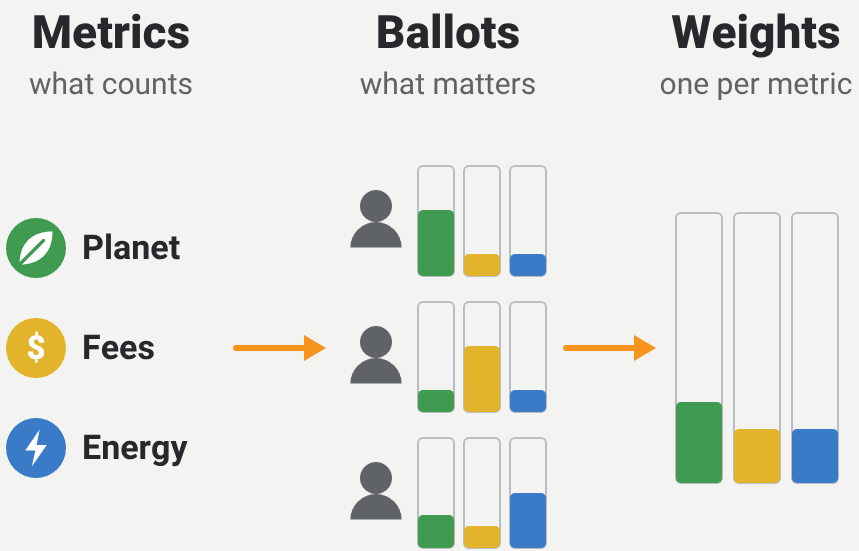
How it works

Metrics
what counts

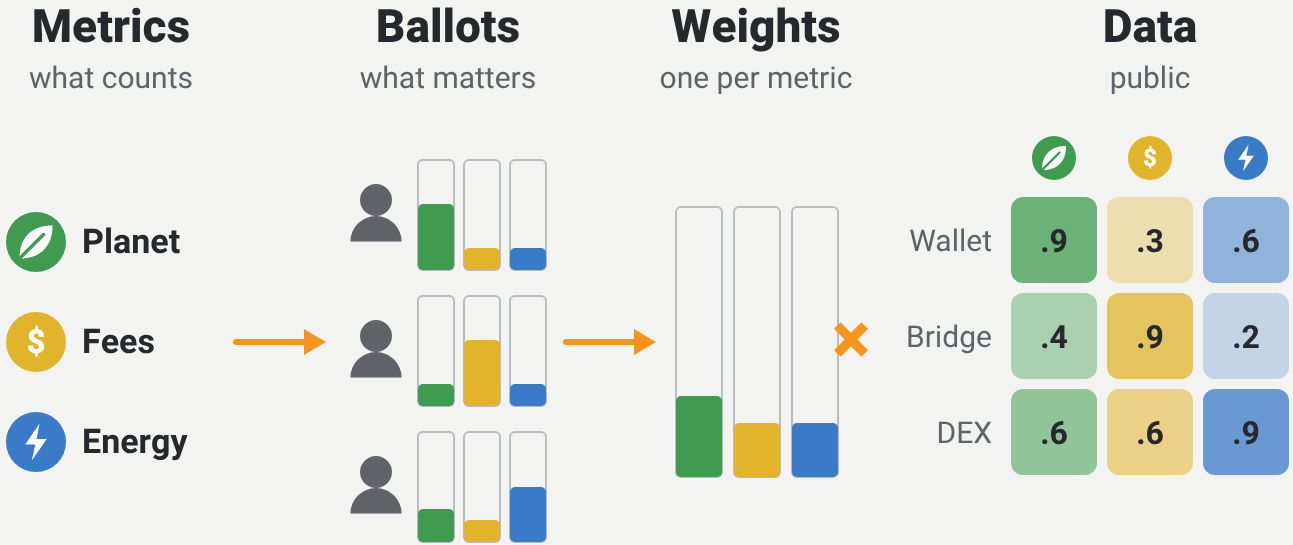
Ballots
what matters



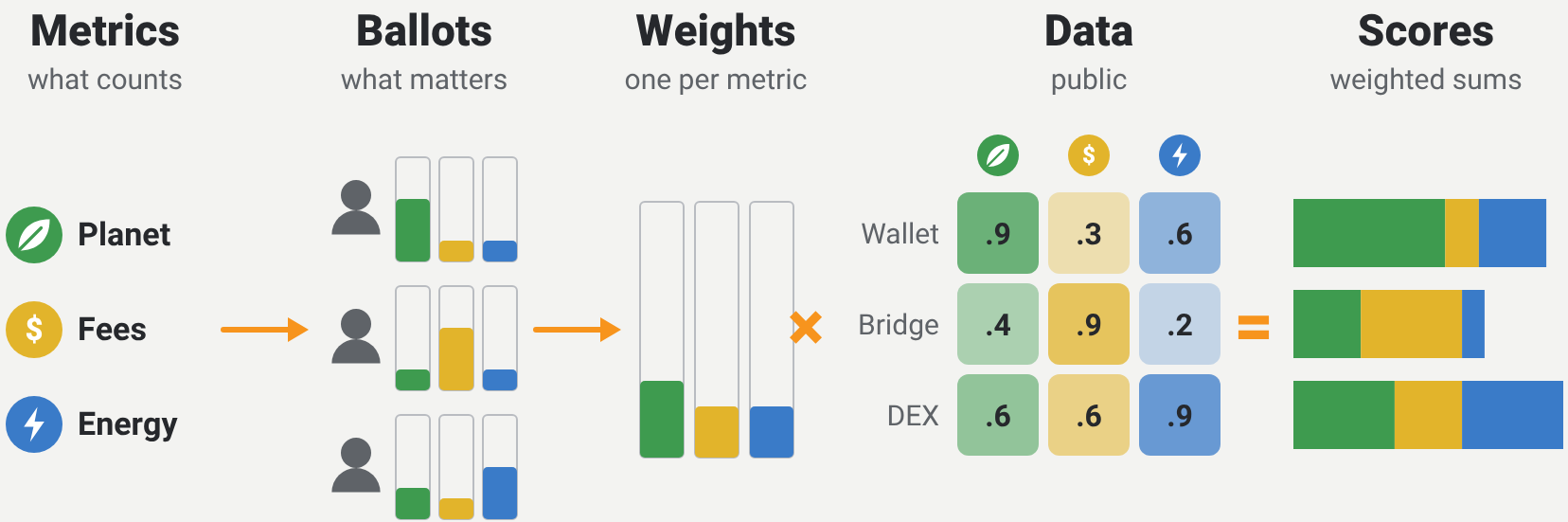
How it works


Metrics
what counts
Ballots
what matters
Weights
one per metric
Planet
Fees
Energy

How it works

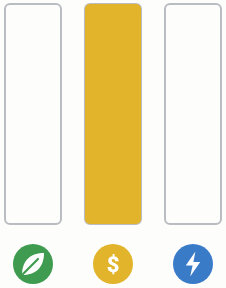

Metrics
what counts
Ballots
what matters
Weights
one per metric
Data
public
Planet
Fees
Energy
$
Wallet .9 .3 .6
Bridge .4 .9 .2
DEX .6 .6 .9

How it works

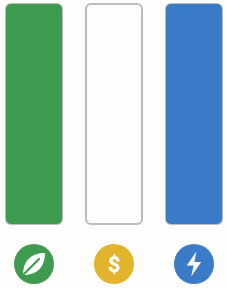

Metrics
what counts
Ballots
what matters
Weights
one per metric
Data
public
Scores
weighted sums
Planet
Fees
Energy
$
Wallet .9 .3 .6
Bridge .4 .9 .2
DEX .6 .6 .9

What a ballot looks like

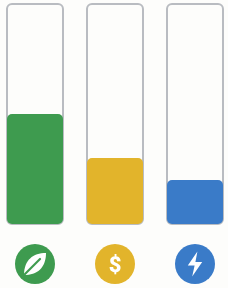
Plurality
pick one



Approval
in or out

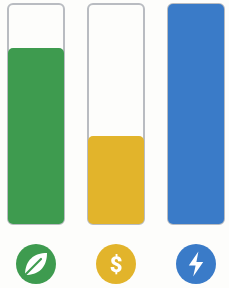


Cumulative
split 100%

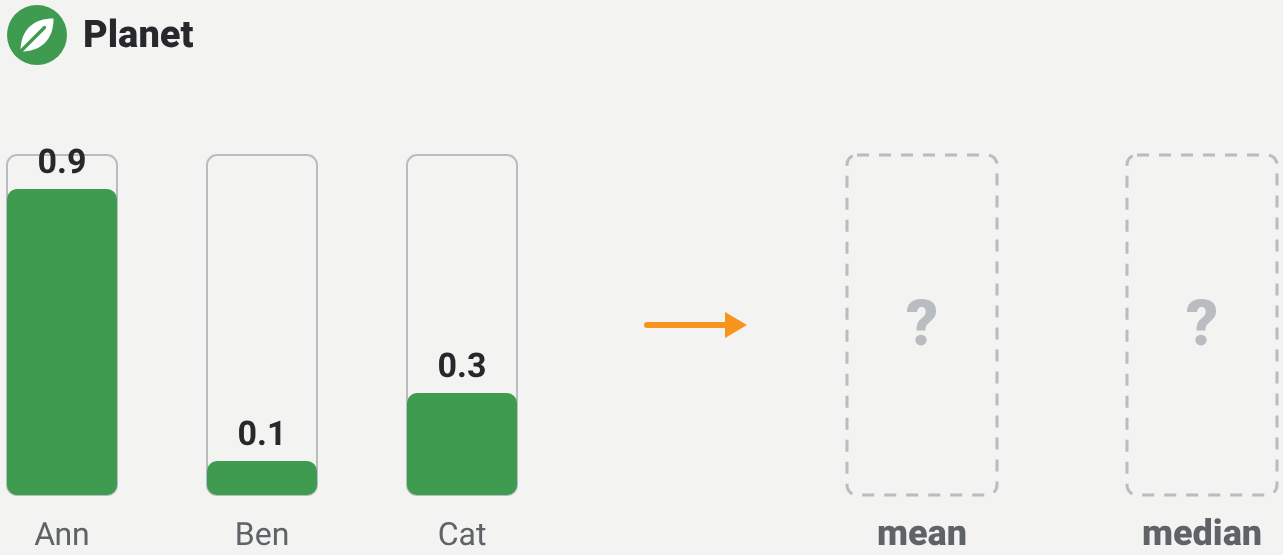


= 100%

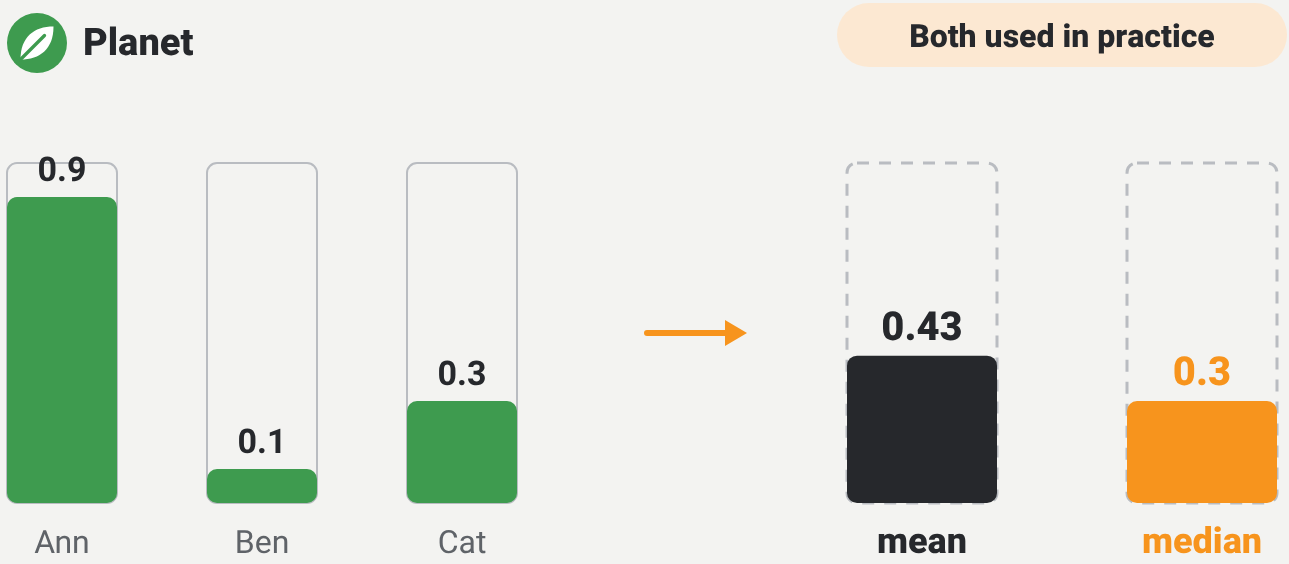
Fractional
any weight



From ballots to one weight: mean or median?



From ballots to one weight: mean or median?



Next: mean vs. median, on three fronts

Lying

Neither is safe

Hitting a target

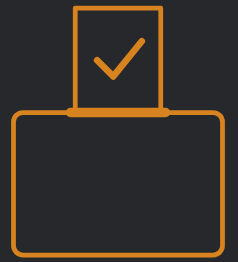
Depends on the ballot

Copying metrics

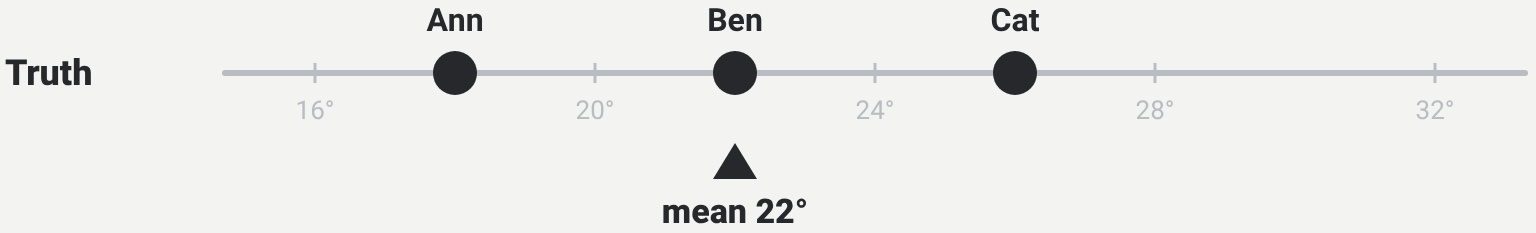
Mean often immune

Strategic voting

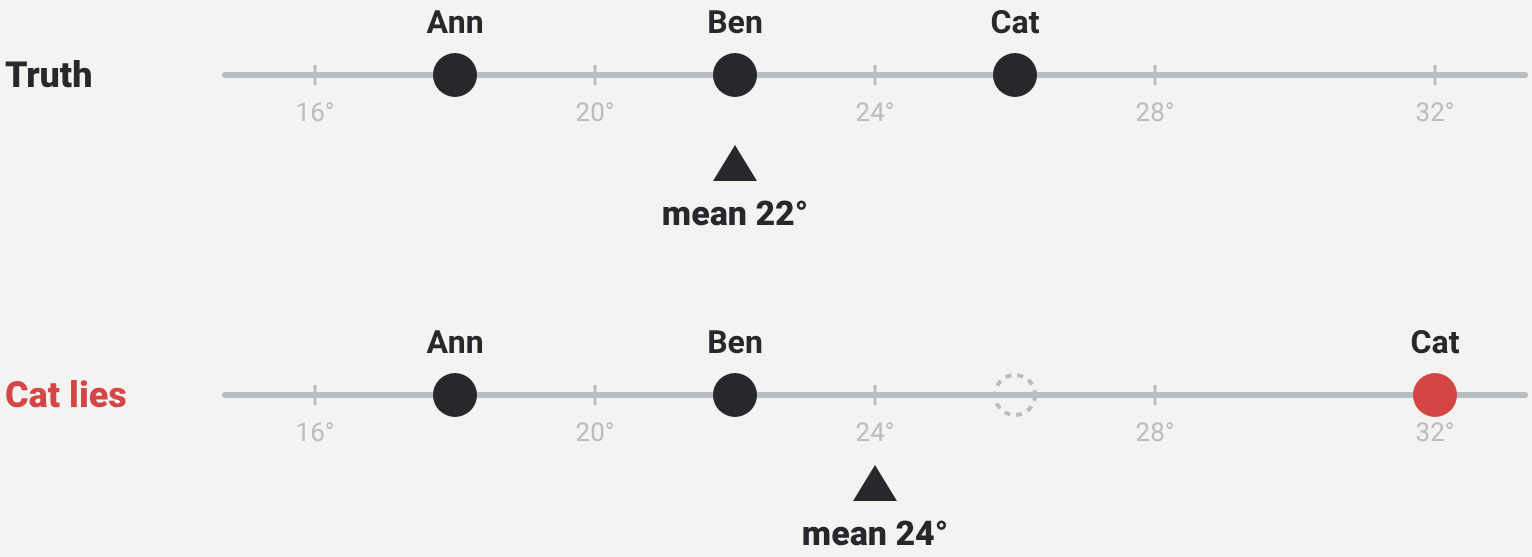
Can a voter gain by lying?



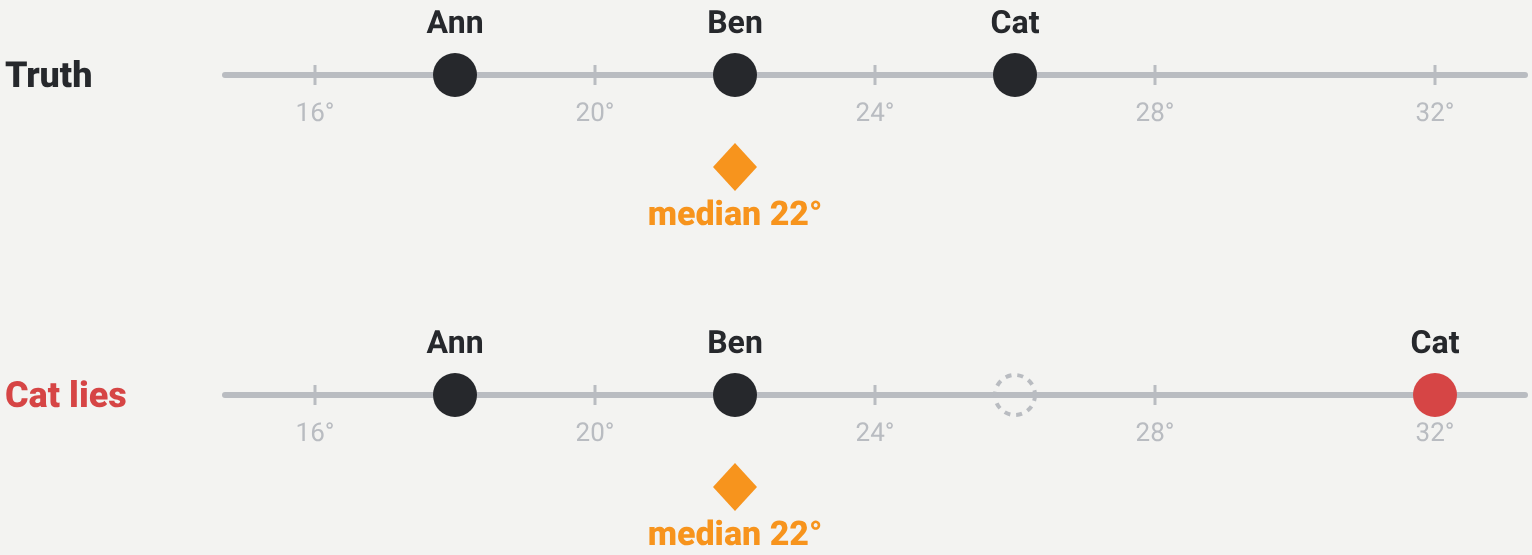
The office thermostat



Cat says 32°: the mean moves



Cat says 32°: the median does not



So a median per metric is safe too?

A project worth 1 on Planet and on Energy

	Planet	Energy	score
Ann (truth)	1	0	**wants 1**
Ben	0	1	
Cat	0	0	
Median	0	0	**0**



Ann

A project worth 1 on Planet and on Energy

	Planet	Energy	score
Ann (lie)	1	1	**wants 1**
Ben	0	1	
Cat	0	0	
Median	0	1	1



Ann

Why

Ann cares about the score, not about each weight

She lost on Planet, so she bought it back on Energy

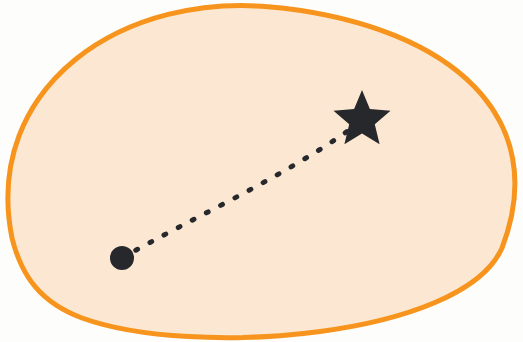
No sensible metric-by-metric rule is safe from this



Lying can pay.
How hard is it to find the lie?

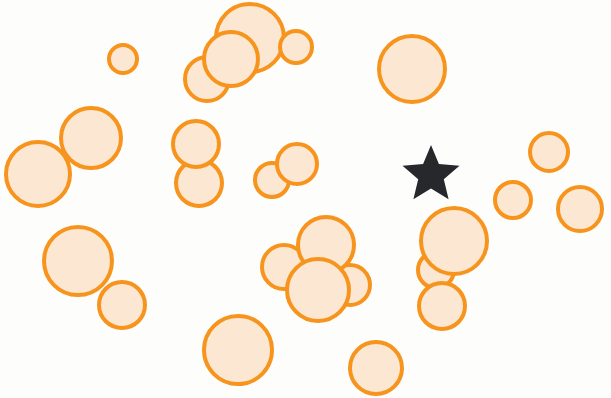
Searching for the right lie

Mean



One smooth region: **easy** to search

Median, cumulative ballots



Scattered islands: a **hard** puzzle

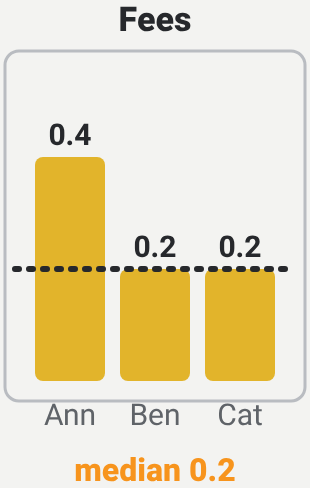
The platform's choices

Duplicate metrics

Round 4: gas fees, and also log gas fees



Three voters, one Fees metric



$ Fees weight

median: 0.2

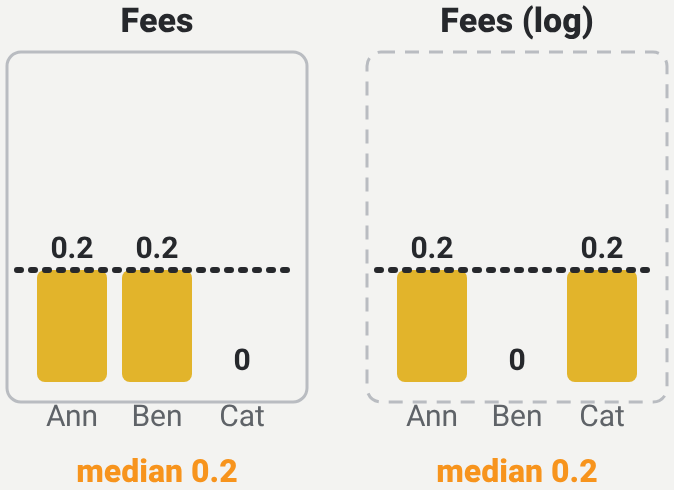
mean: 0.27

DEX, strong on fees:

its score from fees

0.18

Add a copy. Same opinions.



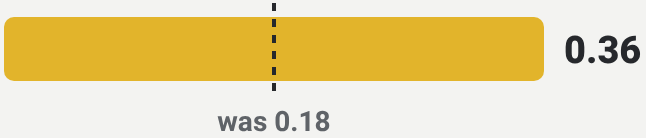
$ Fees weight, in total

median: 0.2 + 0.2 = 0.4

mean: 0.27 (unchanged)

DEX, strong on fees:

its score from fees



The median doubled Fees

Four levers, four ballot types

	Change a ballot		Edit ballots		Pick metrics		Copy metrics	
	mean	median	mean	median	mean	median	mean	median
Plurality	easy	easy	hard	easy	hard	easy	no effect	easy
Approval	hard	hard	hard	hard	hard	hard	hard	hard
Cumulative	easy	hard	easy	hard	hard	hard	no effect	hard
Fractional	easy	easy	easy	easy	hard	hard	hard	hard

Which rule is harder to steer depends on the ballot

Metric-based voting is effective, but not easy to do right

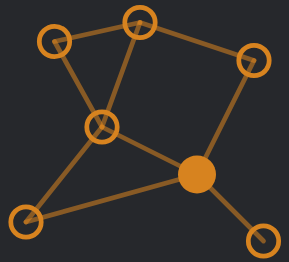


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